

a

Semblance 2.0

Drug recommendations from GDSC2 + CTRP single-agent screening x CCLE tumor-cell profiles.

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 **Cell Line Lookup** — pick a profiled cancer cell line and get its ranked drugs: **broad-spectrum** (what its cancer type is generically most sensitive to) plus **mechanism-specific** (drugs with a validated expression signature in that cancer type). -  **Signature Match** — paste a GSEA program and get the drugs whose drug-sensitive-cell signature it most closely resembles, by biomedical semantic similarity.

 **Research / education only — NOT a clinical tool and NOT medical advice.**

 **Cell Line Lookup**  **Signature Match**

Pick a profiled cell line — get its ranked drug recommendations (broad-spectrum + mechanism).

Cell line
A375 · Melanoma (SKCM)

Precomputed and served straight from the CDN — no backend, always on.

A375 · Melanoma (SKCM) · 648 drugs tested on this line · 53 cell lines of this type

Broad-spectrum — drugs this cancer type is most sensitive to (cancer-type average)

#	DRUG	TARGET	TYPE AVG	N	THIS LINE	IC50 μM
1	Dabrafenib	BRAF	-2.24	62	-5.96	0.0315
2	SB590885	BRAF	-2.19	33	-3.71	1.9185
3	PD0325901	MEK1, MEK2	-1.74	33	-2.65	0.0239
4	ERK_2440	ERK1, ERK2	-1.52	20	-2.71	0.4091
5	ERK_6604	ERK1, ERK2	-1.50	20	-2.53	1.0367
6	THR-103	Mutant RAS	-1.48	21	-0.48	37.5888
7	SCH772984	ERK1, ERK2	-1.43	33	-2.11	0.4043
8	Refametinib	MEK1, MEK2	-1.41	33	-2.46	0.1706
9	Trametinib	MAP2K1, MAP2K2	-1.29	64	-2.67	0.0077
10	Ulixertinib	ERK1, ERK2	-1.23	33	-1.64	1.3569
11	PLX-4032	BRAF	-1.21	45	-0.71	—
12	Selumetinib	MAP2K1, MAP2K2	-1.14	79	-1.37	0.6351
13	MLN2480	ARAF, BRAF, RAF1	-1.07	30	—	—
14	VX-11e	ERK2	-0.98	33	-1.95	1.3903
15	LJ1308	RSK2, RSK1, RSK3	-0.84	33	-0.44	68.5658

Type avg = average sensitivity across all lines of this cancer type (the recommendation), over n lines; This line = this cell line's own value. Both standardized — more negative = more sensitive; IC50 (μM): lower = more potent. Ranks by average potency, so broadly cytotoxic agents can appear high without being lineage-specific — weigh the reported target and n. (Non-therapeutic lab-tool compounds are filtered out.)

Mechanism-specific — drugs with a validated expression signature in this cancer type (exploratory)

DRUG	TARGET	SIGNAL (p)	EVIDENCE (z)
ML162	—	0.71	4.1
1S,3R-RSL-3	GPX4	0.70	3.9
Selumetinib	MAP2K1, MAP2K2	0.64	3.6
PD318088	MAP2K1, MAP2K2	0.56	3.6
GDC-0879	BRAF	0.58	3.6
ML210	—	0.63	3.5
Trametinib	MAP2K1, MAP2K2	0.57	3.3
BI-2536	PLK1	0.49	3.3
PLX-4720	BRAF	0.50	3.2
tacedinaline	HDAC1, HDAC2, HDAC3	0.52	3.2
Merck60	HDAC1, HDAC2	0.50	3.1
topotecan	TOP1	0.44	3.0
SGX-523	MET	0.54	3.0
triazolothiadiazine	PDE4A, PDE4B, PDE4D	0.52	2.9
TG-101348	JAK2	0.48	2.9
CHIR-99021	GSK3B	0.52	2.9
PLX-4032	BRAF	0.42	2.8
Telomerase Inhibitor IX	TERT	0.44	2.7
UNC0638	EHMT1, EHMT2	0.45	2.7
Docetaxel	Microtubule stabiliser	0.48	2.7

All shown clear a permutation test at $p \leq 0.01$. Signal (p) = how well the cancer-type expression signature tracks sensitivity; Evidence (z) = standard deviations above a tissue-matched null (higher = stronger, ranks the list). Hypothesis-generating, not a per-patient prediction — and not corrected for testing thousands of drug/cancer pairs.

b

 **Cell Line Lookup**  **Signature Match**

Bring a GSEA program — get drugs whose sensitive-cell signature it most resembles (mechanism match, exploratory).

Paste a GSEA results table — clusterProfiler / lfgsea schema (ID, Description, NES, pvalue, p.adjust...), CSV or TSV

```
ID,Description,NES,pvalue,p.adjust
REACTION_MITOCHONDRIAL_TRNA_AMINOACYLATION,2.113,0.001,0.01
REACTION_TRNA_PROCESSING,2.085,0.001,0.01
REACTION_MITOCHONDRIAL_TRANSLATION,2.074,0.001,0.01
REACTION_RESPIRATORY_ELECTRON_TRANSPORT,2.064,0.001,0.01
REACTION_RESPONSE_OF_EIF2AK4_GN2_TO_AMINO_ACID_DEFICIENCY,2.06,0.001,0.01
REACTION_ACTIVATION_OF_ATR_IN_RESPONSE_TO_REPLICATION_STRESS,2.059,0.001,0.01
REACTION_TRNA_AMINOACYLATION,2.059,0.001,0.01
```

New here?  Load an example melanoma signature, then hit Search.

...or upload a file

Choose File

No file chosen

Cancer type (optional)
SKCM

Top K
10

Search

35 pathways in your signature

Best drug matches — drugs whose sensitivity signature most resembles yours

#	DRUG	CANCER TYPE	EVIDENCE (Z)	SHARED PATHWAYS
1	istradefylline	Melanoma (SKCM)	2.70	Mitochondrial Translation Respiratory Electron Transport Tma Processing
2	etomoxir	Melanoma (SKCM)	2.47	Mitochondrial Translation Respiratory Electron Transport Tma Aminoacylation
3	GDC-0879	Melanoma (SKCM)	2.10	Respiratory Electron Transport Mitochondrial Translation Dna Replication
4	PD318088	Melanoma (SKCM)	1.99	Mitochondrial Translation Respiratory Electron Transport Activation Of The Pre Replicative Complex
5	SZ4TA2	Melanoma (SKCM)	1.97	Mitochondrial Translation Respiratory Electron Transport Mitochondrial Protein Import
6	A-366	Melanoma (SKCM)	1.84	Respiratory Electron Transport Mitochondrial Translation Tma Aminoacylation
7	MLN2480	Melanoma (SKCM)	1.79	Respiratory Electron Transport Aerobic Respiration And Respiratory Electron Transport Sialic Acid Metabolism
8	Selumetinib	Melanoma (SKCM)	1.69	Dna Replication G2 M Checkpoints Activation Of The Pre Replicative Complex
9	pumorphamine	Melanoma (SKCM)	1.62	Mitochondrial Translation Respiratory Electron Transport Defects In Vitamin And Cofactor Metabolism
10	Ulixertinib	Melanoma (SKCM)	1.62	Activation Of The Pre Replicative Complex Mitochondrial Translation Activation Of Atr In Response To Replication Stress

 Exploratory / hypothesis-generating: semantic similarity of your up-program to drug-sensitive-cell signatures. Not a validated predictor and NOT clinical advice.